Prague University of Economics and Business
Faculty of Informatics and Statistics



A Decomposition of 10-K Risk-Factor Language and Post-Filing Stock Return Volatility

Bachelor Thesis

Study programme: Data Analytics

Author: Šimon Slanský

Supervisor: prof. RNDr. Ing. Michal Černý, Ph.D.

Prague, January 2027

Abstract

Corporate 10-K filings contain both audited financial data and narrative disclosures written by management, yet most empirical models use these two information channels separately. The Item 1A risk-factor section is itself the subject of two parallel literatures which predict effects in opposite directions: one, represented by Campbell et al. (2014), connects a larger *volume* of risk discussion to higher subsequent volatility, while the other, represented by Hope et al. (2016), links a higher *specificity* of risk discussion to lower volatility. This thesis proceeds in three steps. It first separates volume (the log word count of Item 1A) from specificity (the Loughran–McDonald risk-word density) and enters them jointly in a panel regression of post-filing log-volatility. It then splits the composite risk-density into its two sub-lists—uncertainty and litigious wording—each normalised by Item 1A length. Finally, it asks whether what a firm *omits* also matters, measuring for each firm-year the share of risk topics disclosed by industry peers but absent from its own Item 1A. All specifications use two-digit SIC industry and fiscal-year fixed effects with two-way clustered standard errors by firm and fiscal year; the omission specification adds firm fixed effects to absorb permanent disclosure styles. On an unbalanced panel of 526 large U.S. firms over fiscal years 2010–2024 (6,048 firm-years at the 30-day horizon), longer Item 1A sections predict higher post-filing volatility ($\hat{\beta} = +0.116$, $t = +7.26$), while denser Item 1A sections, holding length constant, predict *lower* post-filing volatility ($\hat{\beta} = -4.36$, $t = -3.17$). The decomposition shows that the negative effect comes from the LM-Litigious sub-list ($\hat{\beta}^{\text{lit}} = -6.55$, $t = -3.83$), while the Uncertainty sub-list is not significant. The length effect holds at every horizon from 5 to 365 days, and the litigious effect at every horizon from 10 to 365 days. Both hold in the pre-COVID and COVID-era sub-samples. The density result is a between-firm pattern: under firm fixed effects the density coefficients vanish, so firms that always use more litigious wording have lower volatility, but a firm that raises its litigious wording does not. The peer-relative omission gap is not significant in the cross-section but is positive and significant under firm fixed effects ($\hat{\beta}^{\text{om}} = +1.247$, $t = +2.58$ at the 30-day horizon): firms that skip topics their peers report have higher subsequent volatility. This result is weaker in some robustness checks, so we read it as suggestive.

Keywords

textual analysis, 10-K filings, SEC EDGAR, volatility, risk factors, Item 1A, Loughran–McDonald, LM-Litigious word list, boilerplate disclosure, panel data, corporate disclosures

Contents

Introduction 6

1 Literature Review 7

2 Data 8

2.1 Selecting the Sample . . . 10
2.2 Choosing the Filing Frequency — why annual 10-Ks? . . . 11
2.3 XBRL Tag Consistency Dilemma . . . 13
2.4 How do we extract text from 10-K filings? . . . 14
2.4.1 Source of textual data . . . 14
2.4.2 Section identification and exhibit fallback . . . 15
2.4.3 Post-Processing . . . 16
2.4.4 Coverage and Limitations . . . 16

3 Methodology 18

3.1 Outcome variable: post-filing realised volatility . . . 18
3.2 Control variables . . . 18
3.3 H1 — Item 1A length and disclosure volume . . . 19
3.4 H2 — Risk-word density and disclosure specificity . . . 19
3.4.1 Composite RiskDensity . . . 20
3.4.2 LM sub-list decomposition (second stage) . . . 21
3.5 H3 — Peer-relative omission gap . . . 21
3.6 Estimation and inference . . . 23

4 Results 25

4.1 Descriptive statistics . . . 25
4.2 Volume, specificity, and the LM-list decomposition . . . 27
4.3 Strategic-silence hypothesis and within-firm post-filing volatility . . . 30

5 Robustness 31

5.1 Alternative post-filing windows . . . 31
5.2 Pre-COVID and COVID-era sub-samples . . . 32
5.3 Within-firm identification: firm fixed effects . . . 34
5.4 Robustness of the strategic-silence result (H3) . . . 35
5.5 Stationarity of the panel . . . 38

Conclusion 41

References 42

A Variable Definitions and Data Pipeline 45

A.1 Filing Frequency Diagnostics . . . 45

A.2 XBRL Tag Provenance 47

B Documentation of AI Tool Usage **48**

List of Abbreviations

API Application Programming Interface
CIK Central Index Key
DV Dependent Variable
EDGAR Electronic Data Gathering, Analysis, and Retrieval
EPS Earnings Per Share
FE Fixed Effects
FY Fiscal Year
FYE Fiscal Year End
GLM Generalized Linear Model
L&M Loughran and McDonald
LLM Large Language Model
MD&A Management's Discussion and Analysis
NCI Non-Controlling Interest
NLP Natural Language Processing
OCF Operating Cash Flow
OLS Ordinary Least Squares
ROA Return on Assets
SEC Securities and Exchange Commission
SIC Standard Industrial Classification
TOC Table of Contents
XBRL eXtensible Business Reporting Language

Introduction

1. Literature Review

2. Data

This chapter describes how the firm-year panel is built. The dataset comes entirely from primary sources: no commercial research database (such as Compustat, CRSP, or a vendor-supplied 10-K text corpus) is used. Financial variables come directly from the SEC EDGAR Company Facts API, narrative sections are parsed from the original HTML of the filings, and return-based measures are computed from raw daily prices. This has two advantages. First, every transformation of the data is documented; there are no undocumented vendor adjustments between the filing and the regression input. Second, the textual and financial data can be matched on the same filing, fiscal period, and reporting entity, which matters when the research question concerns the information in a specific document. The cost is that several data problems that commercial providers usually handle (XBRL tag heterogeneity, inconsistent HTML structure, amendments) have to be dealt with directly. Much of this chapter describes how these problems were solved and how the choices were validated.

Figure 2.1 gives an overview of the data-preparation process. The pipeline starts from the raw SEC sources, applies the sample filters, and then runs two parallel branches—a financial (XBRL) branch and a textual (10-K narrative) branch—that are merged into the final regression-ready firm-year panel.

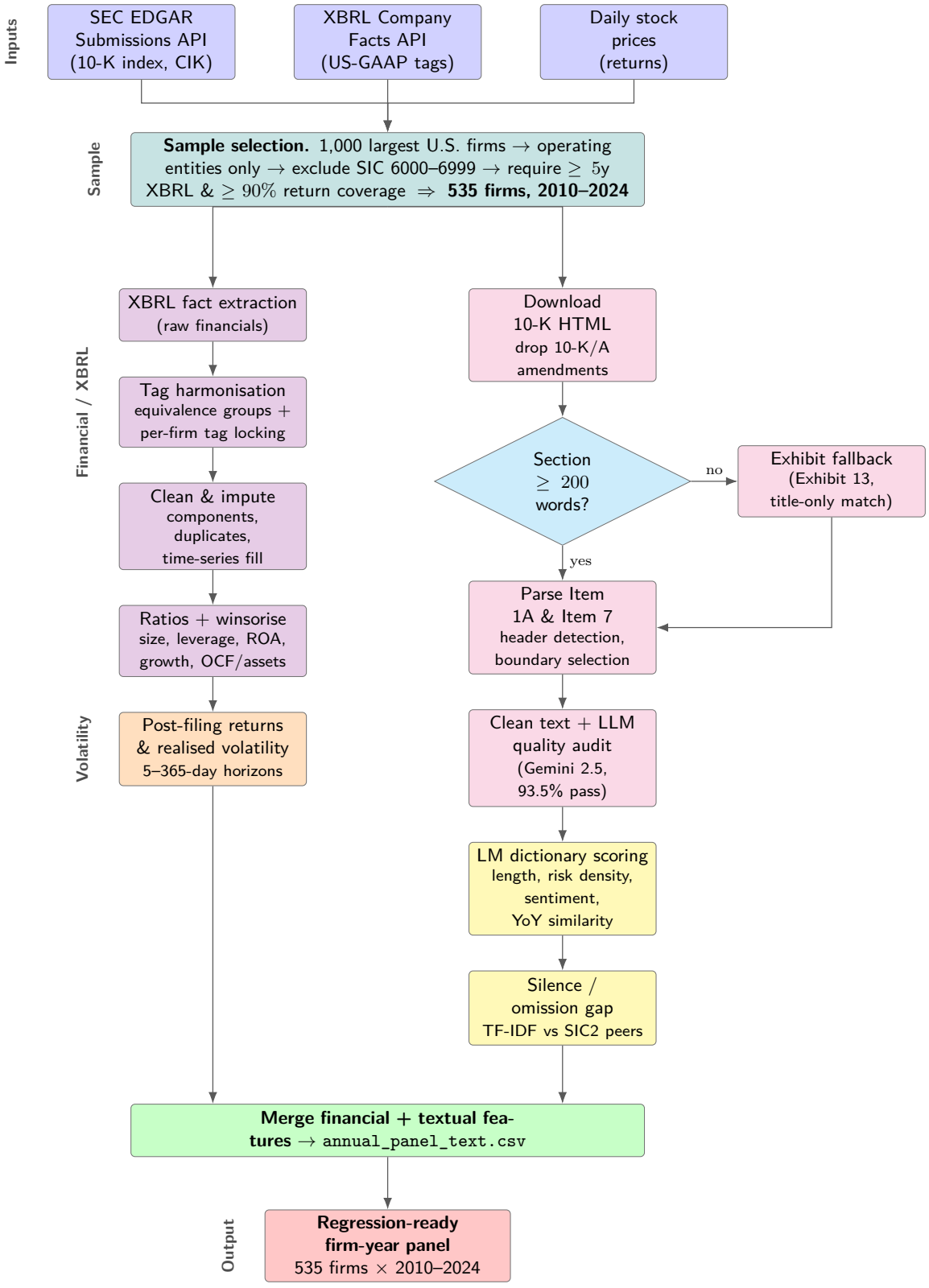


Figure 2.1: Operational pipeline for the complete data preparation. The workflow runs top-to-bottom from the raw SEC sources through sample selection, then splits into a financial (XBRL) branch and a textual (10-K narrative) branch that are merged into the final regression-ready firm-year panel. Own processing.

2.1 Selecting the Sample

The starting universe is the set of the 1,000 largest U.S. public companies by market capitalisation, identified from the SEC EDGAR company-tickers file, which maps each ticker to its Central Index Key (CIK) and exchange listing. Three filters are then applied in sequence. First, we keep only entities that the SEC classifies as "operating" companies. This step removes exchange-traded funds, closed-end funds, blank-cheque (SPAC) vehicles, and shell entities, none of which produce the kind of narrative risk-factor disclosure that the thesis studies, and whose financial statements are not comparable to those of operating businesses. Second, we exclude firms with SIC codes in the range 6000–6999 (finance, insurance, and real estate), as is standard: financial firms face different regulation, their balance sheets consist mostly of financial assets and liabilities, and ratios such as leverage or operating cash flow over assets mean something different for them. Third, we require at least five fiscal years of readable XBRL data from 10-K filings, and we keep a firm only if at least 90% of its filings have a valid 365-day post-filing volatility. A filing has a valid volatility when the stock has at least 200 daily prices in the year after the filing. The five-year minimum gives enough within-firm variation for fixed-effects estimation. The return rule removes firms with long trading gaps. Because this rule looks at prices *after* each filing, it also removes some firms that were delisted during the sample; together with the fact that the starting list contains firms that exist today, this means the sample favours surviving firms (see the limitations in Section 3.6).

Sample size. We focus on large, liquid firms rather than the full EDGAR universe for several reasons. First, the regressions use firm fixed effects with standard errors clustered by firm and fiscal year. As Petersen (2009) emphasises, the reliability of clustered inference depends mainly on the number of clusters, not the number of observations. With 526 firm clusters, the cross-section is large enough for cluster-robust inference. Second, large firms have more complete machine-readable filing histories, which means fewer missing observations and less measurement error in the text pipeline. Third, smaller firms are more likely to be illiquid, thinly traded, or to have poorly formatted filings, which adds noise to both return-based and accounting variables. Finally, a large-firm sample keeps the text processing manageable and gives a relatively homogeneous set of issuers.

The sample size is in line with prior work on the text and financials of annual filings. Intermediate stages of the pipeline contain between 539 and 548 firms; the 30-day regression sample contains 526 firms. Campbell et al. (2014) study about 9,000 firm-years of risk-factor disclosures from large U.S. firms; Loughran and McDonald (2011) use a broader set of roughly 50,000 firm-years, but their goal—validating a sentiment dictionary—calls for maximal cross-sectional coverage, whereas the joint volume–specificity test here uses within-industry-year variation across about 526 firms.

2.2 Choosing the Filing Frequency — why annual 10-Ks?

We use annual 10-K filings rather than quarterly 10-Q reports. Quarterly data would roughly triple the sample, but two practical problems, shown in Tables 2.1–2.2, rule this out.

Financial data gaps in Q4. 10-Q filings cover only Q1–Q3, so any quarterly panel must reconstruct Q4 as the difference between the annual figure and the sum of the three preceding quarters. Table 2.1 shows that this residual approach is not usable for operating cash flow: the three quarterly facts coexist in only 7% of firm-years, because most firms report cumulative year-to-date cash flows rather than standalone quarterly amounts. Operating cash flow is the primary input into our OCF-to-assets ratio, so a quarterly panel would lose the cash-flow-based robustness check at the source. Even for net income and operating income, where coverage exceeds 80%, the implied Q4 disagrees with the explicit Q4 by more than one percent of the annual total in about 10–14% of comparable firm-years.

The residual and the reported fourth quarter differ because they come from documents prepared at different times and with different scrutiny. Interim 10-Q figures are reviewed but not audited. The 10-K is audited, and year-end audit adjustments (impairments, tax true-ups, changes in estimates, reclassifications to discontinued operations) are usually booked in the fourth quarter without restating earlier quarters. The residual Q4 absorbs these adjustments, whereas an explicit Q4 disclosed in the 10-K (for example in a selected-quarterly-data note) may have been recast on a comparable basis. The explicit Q4 value in the Company Facts API also often comes from a *later* filing that shows the prior period as a restated comparative, so the two values can reflect different versions of the same accounting history. 52/53-week fiscal calendars and rounding or scaling differences add small mechanical gaps. None of these is a data error as such, but a reconstructed quarterly panel would mix audited and unaudited, restated and original figures in ways that are hard to control.

Table 2.1: Q4 Reliability: Availability and Consistency of Fourth-Quarter XBRL Facts

Metric	Firm-Years	Q1 – Q3 (%)	Explicit Q4 (%)	Comparable	Match (%)
Net Income	451	80.9	43.0	172	89.5
Operating Income	391	79.0	22.0	72	86.1
Operating Cash Flow	462	7.4	3.2	0	—

Note: "Q1 – Q3" indicates the share of fiscal years where all three quarterly facts (Q1, Q2, Q3) exist in XBRL. "Explicit Q4" indicates the share with a standalone fourth-quarter fact (period ≈90 days ending at fiscal year-end). "Comparable" is the count of firm-years for which both an explicit Q4 fact and the implied residual are available. "Match" reports the share of those comparable observations where the two values agree within 1% of the annual figure. Sample: 30 non-financial firms, all available fiscal years.

Missing risk-factor text in 10-Q. SEC Regulation S-K only requires firms to update risk factors in quarterly filings when "material changes" have occurred since the last 10-K. In

practice, only about half the 10-Q filings in our pilot sample (15 firms, 90 filings) contained an actual risk-factor section. The rest only referred back to the annual report (Table 2.2). The year-on-year text variables in our specifications therefore cannot be built at a quarterly frequency.

Table 2.2: Textual Data Availability: 10-K vs. 10-Q Filings

	10-K	10-Q
Filings tested	44	90
MD&A (Management's Discussion and Analysis)		
Section found (%)	93	100
Median word count	20,494	5,806
Risk Factors		
Substantive section found (%)	100	52
Boilerplate / reference only (%)	0	48
Median word count (substantive)	29,902	9,144

Note: "Found" requires >200 words extracted by regex-based section parser. "Substantive" for 10-Q excludes filings containing only a reference to the annual 10-K (e.g., "no material changes from those disclosed in our Annual Report"). MD&A corresponds to Item 7 in 10-K and Part I Item 2 in 10-Q; Risk Factors corresponds to Item 1A in 10-K and Part II Item 1A in 10-Q. Sample: 15 firms, most recent filings.

Chosen frequency. Tables 2.1 and 2.2 favour an annual design. A quarterly panel would require rebuilding fourth-quarter accounting variables from incomplete and often inconsistent XBRL data, and would rely on risk-factor sections that are often missing from 10-Q filings because firms need only report material changes since the last annual report. Both problems reduce data availability and add measurement error. Annual 10-K filings contain full fiscal-year accounting data and complete MD&A and risk-factor sections, so all explanatory variables can be measured the same way across firms and years.

The annual frequency also matches the disclosure process. The text measures in this thesis describe firms' full annual filings, not quarterly updates. Matching the observation interval to the reporting frequency of the documents avoids imputation and artificial within-year variation, and changes in the text measures then correspond to actual revisions of the annual disclosure.

An annual firm-year panel is also the convention in the 10-K textual-analysis literature. Li (2008) analyses 55,719 firm-year observations constructed from annual 10-K filings, while Loughran and McDonald (2011) develop and validate financial text dictionaries using a sample of 50,115 firm-year 10-K filings. Both studies treat the annual filing as the unit of analysis. The XBRL coverage comparison (Table A.1) and a consolidated summary (Table A.2) are in Appendix A.

2.3 XBRL Tag Consistency Dilemma

In the SEC EDGAR Company Facts API, one accounting concept can be reported under several US-GAAP taxonomy elements. Filers have some discretion in choosing concepts, and the taxonomy changes over time. Prior XBRL research shows that the same accounting variable may map to several taxonomy concepts, so these must be mapped explicitly before analysis (Chychyla & Kogan, 2015). For example, operating cash flow may be reported as either `NetCashProvidedByUsedInOperatingActivities` or `NetCashProvidedByUsedInOperatingActivitiesContinuingOperations`; net income may be reported as `NetIncomeLoss`, `ProfitLoss`, or `IncomeLossFromContinuingOperations`; and stockholders' equity may be presented either including or excluding non-controlling interests (NCI). These tags often overlap but are not always identical. Treating all of them as interchangeable would add measurement noise, while choosing one tag for all firms would drop observations for firms that consistently use another valid element.

We handle this problem with a two-part *tag-harmonisation* process: (i) *equivalence groups* for tags that are economically interchangeable, and (ii) *per-firm tag locking* for tags that are not.

Equivalence groups. Following the concept-mapping approach of Chychyla and Kogan (2015), we define equivalence groups for tags that differ in label or taxonomy representation but not in economic content. `NetCashProvidedByUsedInOperatingActivities` and `NetCashProvidedByUsedInOperatingActivitiesContinuingOperations` are treated as equivalent operating-cash-flow measures, while `NetIncomeLoss` and `ProfitLoss` are treated as interchangeable net-income measures. By contrast, `IncomeLossFromContinuingOperations` is excluded from the equivalence set because it excludes discontinued operations and so measures something different.

We also check these groupings in the data. In the Company Facts API there are 6,113 firm-years in which both members of an equivalence pair are reported. The differences between paired tags are negligible, with a median ROA bias of 0.033 percentage points (Appendix A, Table A.4), so we treat the paired tags as two representations of the same quantity. In the few cases where both tags are present and differ materially, the parent-company value is used.

Per-firm tag locking. Where alternative tags measure different things, such as stockholders' equity with versus without NCI, or `IncomeLossFromContinuingOperations` versus total net income, mixing tags within a firm's time series would create measurement error. We assign each firm the tag it uses most often over its reporting history and set values reported under other tags to missing.

Missing balance-sheet values. If total liabilities are missing, we compute them as total assets minus stockholders' equity. If total assets or total liabilities are still missing, we copy

the value from the firm's previous fiscal year, but only when that year is directly before. We never fill a value from a *later* year, because a later balance sheet was not public at the time of the filing.

With both procedures, the complete-case rate for the five regression controls rises from 78.9% to 92.7% of firm-years. A small inconsistency remains: about 4% of firms use a continuing-operations operating-cash-flow tag while reporting total net income. Because these mismatches are small, we keep the observations and rely on winsorisation to limit their influence. Such researcher-imposed harmonisation is common in XBRL-based panel work (Chychyla & Kogan, 2015).

Per-firm tag choices and the overlap diagnostic are reported in Appendix A (Table A.4).

2.4 How do we extract text from 10-K filings?

The study uses two sections of the 10-K: Item 1A *Risk Factors* and Item 7 *Management's Discussion and Analysis of Financial Condition and Results of Operations* (MD&A). This section describes how we locate and extract them from the raw HTML filings on SEC EDGAR.

2.4.1 Source of textual data

All filings are retrieved directly from SEC EDGAR, the Commission's public repository of corporate disclosures, without any intermediate vendor or pre-processed corpus. Retrieval proceeds in two steps. First, the EDGAR Submissions API[1] is queried for each company's Central Index Key (CIK). The response lists every filing the company has ever submitted together with its form type, filing date, accession number, and the name of the *primary document*, i.e. the main HTML file of the filing as opposed to its exhibits. Second, for each original 10-K the primary document is downloaded from the EDGAR archive at `sec.gov/Archives/edgar/data/{CIK}/{accession}/{document}`. Where the filing index indicates that narrative content has been placed in an exhibit (Section 2.4.2), the exhibit file is fetched from the same accession folder. All requests carry the declared user-agent header that the SEC requires and are throttled to stay below the published rate limit of ten requests per second. The download can be reproduced from the scripts in the accompanying repository.

From the panel of about 6,700 firm-year observations, we drop 10-K/A amendments. These typically consist of a cover page and a restated exhibit and do not contain the full narrative sections of the original 10-K. Financial data for amended fiscal years are kept, since the XBRL

[1] `https://data.sec.gov/submissions/CIK{cik}.json`; the endpoint family is documented at https://www.sec.gov/search-filings/edgar-application-programming-interfaces. The same endpoint family provides the Company Facts data used for the financial branch (Section 2.3).

pipeline already picks up amended figures when available (Section 2.3). Text is extracted only from original 10-K filings.

2.4.2 Section identification and exhibit fallback

Unlike structured XBRL data, the textual parts of 10-K filings are buried inside HTML with no standardised tagging, mixed with table-of-contents (TOC) entries, running headers, and inline cross-references. A naive search for the string "Item 1A" would therefore typically land on the TOC entry or on a cross-reference such as "see Item 1A" rather than on the section heading. We use a four-stage HTML parser that operates on the document structure *before* the text is flattened:

1. **Candidate detection.** The parser scans the HTML tree for elements whose text matches a regular expression for the target "Item X" strings ("Item 1A", "Item 7", and the headers that delimit them). A match is kept as a candidate header only if it also carries at least one formatting cue typical of a section heading: bold font weight, ALL-CAPS text, or a font size of at least 12 pt. Two classes of false positives are rejected outright: elements that are hyperlinks (navigation links in the TOC) and elements that are surrounded by other "Item" references within a window of 300 characters, which is the characteristic signature of a table of contents.
2. **In-place tagging.** Each confirmed candidate is marked with a unique sentinel string inside the HTML tree. Only then is the document converted to plain text. Tagging before flattening preserves the position of the header exactly, whereas locating headers in already-flattened text would lose the formatting information used in step 1.
3. **Candidate selection.** When several candidates for the same Item survive (for instance a TOC entry that happened to be bold), the parser selects the one followed by the longest stretch of text before the next expected Item header. The true section is, by construction, the one with substantive content after it.
4. **Section bounding.** The section ends at the next Item header in filing order: Item 1B, 1C, or 2 for Item 1A, and Item 7A or 8 for Item 7. Accepting several terminators handles filings that omit optional items (Item 1B is not required for smaller filers; Item 1C was only introduced for fiscal years ending after December 2023).

Exhibit fallback. Some filers, especially in the early part of the sample, incorporate the narrative sections by reference to the annual report to shareholders, which is attached as Exhibit 13. In such cases the main document contains only a short pointer ("The information required by this Item is incorporated by reference to . . . "). If the recovered section is shorter than 200 words, the parser treats it as a stub, follows the reference to the exhibit file listed in the filing index, and reruns the same four-stage procedure on the exhibit using title-only patterns ("Risk Factors", "Management's Discussion and Analysis"), since exhibits do not carry Item numbering.

2.4.3 Post-Processing

After HTML-to-text conversion the parser strips three layout artefacts: "⟨*page number*⟩ *Table of Contents*" headers, standalone page numbers on lines that do not contain any other text, and non-breaking-space characters. The cleaning rules target only patterns that cannot be part of meaningful text and are applied after section boundaries have already been set.

2.4.4 Coverage and Limitations

The extraction pipeline was validated on two samples: the 50 most recent filings (fiscal year 2024, both sections recovered in all 50) and a stratified random sample of 100 filings covering 2010–2024 (97/100 successful, with three failures driven by PDF-only exhibits or the absence of any formatting cue for the section header). On the full panel we expect a coverage rate of roughly 97%, with missing values treated as missing in the textual variable construction.

LLM-assisted quality audit. To check that the parser finds and bounds the right sections, we had an independent reviewer assess a stratified random sample of filings. Independent annotation is a standard way to assess automated text processing; Benoit et al. (2016) show that reliable aggregate assessments can be obtained even when individual judgments are imperfect. Large language models (LLMs) perform structured annotation tasks about as well as, and sometimes better than, human annotators, and do so consistently (Gilardi et al., 2023). The LLM was used only as a reviewer of the output; all section boundaries come from the deterministic pipeline described above.

A stratified random sample of 100 filings was drawn, with 20 filings selected from each five-year period between 2010 and 2024. Both Item 1A (Risk Factors) and Item 7 (MD&A) were evaluated for every filing, yielding 200 section-level assessments. This sample size implies Wilson 95 % confidence intervals of approximately ±3 percentage points for proportions near 95 % (Agresti & Coull, 1998).

For each extracted section, Google Gemini 2.5 Flash evaluated five criteria: (i) correct section identity, (ii) start-boundary accuracy, (iii) end-boundary accuracy, (iv) content quality, and (v) completeness. Each criterion was rated as Pass, Minor issue, or Fail. Detailed results are reported in Appendix A (Table B.1). Across all individual criteria, the lenient acceptance rate (Pass + Minor issue) exceeds 95 %. When all five criteria are considered jointly, 187 of 200 extracted sections satisfy the acceptance standard, corresponding to an overall success rate of 93.5 % (95 % CI: 89.2–96.2 %).

Failures are concentrated in older filings: five of the six occur in 2010–2012 and one in 2022–2024, consistent with SEC filing formats becoming more standardised over time. The extraction is accurate, and the remaining errors are mostly in early filings with less standard document structure.

Relation to prior approaches. Our extraction strategy is motivated by Dyer et al. (2017), who document large changes in 10-K disclosures over time and the problems these cause for large-sample textual analysis. Compared with the plain-text extraction used in earlier studies (e.g. Campbell et al., 2014; Li, 2008; Loughran & McDonald, 2011), the HTML-based parser produces fewer false positives from TOC entries, navigation elements, and inline cross-references. Unlike approaches based on XBRL-tagged text blocks (e.g. `us-gaap:RiskFactorsTextBlock`), it also covers a much longer period: these tags became more reliable after the adoption of Inline XBRL but are incomplete and inconsistent in earlier filings. Extracting sections from the HTML lets us apply the same method over the full 2010–2024 sample.

3. Methodology

This chapter sets out the regression specifications for the three hypotheses. All tests use a dynamic panel regression in which post-filing realised volatility depends on its own lag, financial controls, and one or more characteristics of the Item 1A risk-factor section, with industry and fiscal-year fixed effects. Section 3.1 defines the dependent variable; Section 3.2 lists the controls, which are the same in all specifications so that the text coefficients can be compared. Sections 3.3 and 3.4 describe the joint test of the volume channel (H1) and the specificity channel (H2), which must be estimated together because the density measure for H2 has section length in its denominator. Section 3.5 introduces the peer-relative omission gap for H3, and the rest of the chapter covers estimation, inference, and robustness checks.

3.1 Outcome variable: post-filing realised volatility

The dependent variable is the natural logarithm of annualised post-filing stock-return volatility measured over a window of h calendar days:

$$\ln \sigma^{[h]}_{i,t+1} \;=\; \ln\!\left(\sqrt{252}\cdot \mathrm{sd}\big(r_{i,d}\big)_{d\in W^{[h]}_{i,t}}\right), \tag{3.1}$$

where

$$r_{i,d} \;=\; \ln\!\left(\frac{P_{i,d}}{P_{i,d-1}}\right)$$

is the daily log return for firm i, and $W^{[h]}_{i,t}$ denotes the post-filing window associated with filing year t. Day 0 is the first trading day on or after the 10-K filing date. The window begins two trading days after day 0, so the market reaction on the filing day and the next day is left out, and it then runs for h calendar days. Only returns between two prices inside the window are used. A window needs a minimum number of daily returns (3 for $h = 5$, 5 for $h = 10$, 14 for $h = 30$, about half the trading days for longer windows); if it has fewer, or if all prices in it are the same, the volatility is treated as missing. The daily standard deviation is annualised by multiplying by $\sqrt{252}$; this is a constant rescaling and does not affect inference.

We use the natural logarithm of volatility, so a coefficient β implies a change of $100(e^{\beta}-1)\%$ in volatility, approximately $100\beta\%$ for small β. The main specification uses a 30-day post-filing horizon ($h = 30$); horizons of $h \in \{5, 10, 90, 180, 365\}$ are reported as exploratory checks.

3.2 Control variables

Every regression includes the same vector $\mathbf{X}^{\mathrm{fin}}_{i,t}$ of firm-level controls drawn from the annual XBRL filings (Section 2.3): firm size ln(Total Assets), leverage (total liabilities over total assets), return on assets (net income over total assets), and asset growth (year-on-year change

in total assets divided by lagged total assets). Lagged log-volatility $\ln \sigma_{i,t}^{[h]}$ enters with its own coefficient ϕ. It uses the window of the same length h after the firm's filing for the *previous* fiscal year $(t-1)$. Each firm's first fiscal year is not in the regression sample, because asset growth needs the previous year's total assets; that first filing is still used to compute the lagged volatility of the second year. If the firm has no filing for year $t-1$, the lag is missing; we never use an older year instead. The lagged window is also cut off on the day before the current filing, so it never overlaps the outcome window. This matters only for $h = 365$ days, where a full year after the previous filing would otherwise run past the current filing date. All ratios are winsorised at the 1st and 99th percentiles of the regression sample. The H3 regression replaces asset growth with two liquidity proxies (current ratio and operating cash flow over assets). The reason is given in Section 3.5.

3.3 H1 — Item 1A length and disclosure volume

Following Campbell et al. (2014), disclosure volume is measured as the logarithm of the Item 1A word count. Campbell et al. show that firms with longer risk-factor disclosures later have higher market-based risk measures, including return volatility, consistent with disclosure volume carrying information about firm risk. We predict a positive association between Item 1A length and post-filing realised volatility. The volume variable is

$$\text{Words}_{i,t}^{1A} \;=\; \text{number of tokens in Item 1A of firm } i\text{'s year-}t \text{ 10-K}, \tag{3.2}$$

and $\ln(\text{Words}_{i,t}^{1A})$ enters the regression. The text-cleaning pipeline (Section 2.4.3) removes XBRL tags, footnote markers, and table cells before tokenisation, so the count reflects narrative content only. The median Item 1A holds about 8,800 words, with a long right tail (Section 4.1).

The volume channel is identified jointly with the specificity channel of H2 in a single regression (the *joint specification*):

$$\begin{aligned}\ln \sigma_{i,t+1}^{[h]} \;=\; \alpha \;+\; \phi \ln \sigma_{i,t}^{[h]} \;+\; \boldsymbol{\beta}'\mathbf{X}_{i,t}^{\text{fin}} \;+\; \beta^{\text{len}} \cdot \ln(\text{Words}_{i,t}^{1A}) \\ +\; \beta^{\text{dens}} \cdot \text{RiskDensity}_{i,t} \;+\; \mu_j \;+\; \tau_t \;+\; \varepsilon_{i,t},\end{aligned} \tag{3.3}$$

where μ_j is a two-digit SIC industry fixed effect, τ_t is a fiscal-year fixed effect, and RiskDensity is defined in Section 3.4. H1 predicts $\beta^{\text{len}} > 0$. Length and density must enter together because the dictionary risk-density used in prior work has length in its denominator.

3.4 H2 — Risk-word density and disclosure specificity

H2 is motivated by the disclosure-specificity framework of Hope et al. (2016). Hope et al. argue that risk disclosures are more informative when they contain firm-specific details rather

than generic statements that could apply to any firm. They find that more specific risk-factor disclosures lead to stronger market reactions and better analyst assessments of firm risk.

We do not measure specificity directly. Instead, we test whether the word composition of Item 1A is associated with future volatility after controlling for length and the usual financial determinants of volatility. We focus on the share of words from the uncertainty and litigious dictionaries of Loughran and McDonald (2011), which were built for financial reporting.

Two disclosures of the same length may convey information differently. One may use more standard risk vocabulary; the other may use more firm-specific narrative. A higher concentration of dictionary risk words may therefore mean that more of the disclosure is generic risk framing rather than discussion of firm-specific exposures. This interpretation needs care, because words such as "litigation", "regulatory", "investigation", or "compliance" can describe very firm-specific circumstances. Risk-word density is thus an indirect proxy for the balance between standard risk terminology and other narrative, not a direct measure of specificity in the sense of Hope et al. (2016).

Under this interpretation, a section with less standard risk vocabulary leaves more room for firm-specific discussion of risks, operations, counterparties, products, and events. If firm-specific disclosure helps investors understand risk (Hope et al., 2016), higher risk-word density should be associated with lower subsequent volatility, holding length fixed. H2 predicts a negative relation. The test has two stages: the first uses a composite risk-word density, and the second splits it into the two Loughran and McDonald (2011) sub-lists.

3.4.1 Composite RiskDensity

The first-stage variable is the share of Item 1A words that match either the Loughran–McDonald uncertainty or litigious dictionaries:

$$\text{RiskDensity}_{i,t} = \frac{n_{i,t}^{\text{unc}} + n_{i,t}^{\text{lit}}}{\text{Words}_{i,t}^{1A}}, \tag{3.4}$$

where n^{unc} and n^{lit} denote the numbers of uncertainty- and litigious-dictionary words appearing in firm i's Item 1A disclosure and Words^{1A} is the total number of words in that section.

Higher values mean that more of the section consists of dictionary risk words. As discussed above, RiskDensity is not a direct measure of specificity; it measures how prominent standard risk language is in the section. Lower values mean that more of the text falls outside the risk lexicon and may contain more firm-specific information, though not necessarily in every case.

H2 is tested through the coefficient β^{dens} in equation (3.3), with the prediction

$$\beta^{\text{dens}} < 0.$$

3.4.2 LM sub-list decomposition (second stage)

RiskDensity combines two different kinds of risk language. To see which one drives any relation, the second stage splits RiskDensity into two sub-densities:

$$\text{UncDensity}_{i,t} \;=\; \frac{n^{\text{unc}}_{i,t}}{\text{Words}^{1A}_{i,t}}, \tag{3.5}$$

$$\text{LitDensity}_{i,t} \;=\; \frac{n^{\text{lit}}_{i,t}}{\text{Words}^{1A}_{i,t}}, \tag{3.6}$$

such that

$$\text{RiskDensity}_{i,t} \;=\; \text{UncDensity}_{i,t} + \text{LitDensity}_{i,t}.$$

The uncertainty dictionary covers language about ambiguity and incomplete knowledge of future outcomes. The litigious dictionary covers legal exposure, regulation, compliance, investigations, claims, and disputes. Both are risk-related, but they measure different things (Loughran & McDonald, 2011).

The second-stage specification replaces the composite measure with its two components:

$$\begin{aligned}\ln \sigma^{[h]}_{i,t+1} \;=\; \alpha \;+\; \phi \ln \sigma^{[h]}_{i,t} \;+\; \boldsymbol{\beta}'\mathbf{X}^{\text{fin}}_{i,t} \;+\; \beta^{\text{len}} \cdot \ln(\text{Words}^{1A}_{i,t}) \\ +\; \beta^{\text{unc}} \cdot \text{UncDensity}_{i,t} \;+\; \beta^{\text{lit}} \cdot \text{LitDensity}_{i,t} \;+\; \mu_j \;+\; \tau_t \;+\; \varepsilon_{i,t}.\end{aligned} \tag{3.7}$$

The two coefficients are separately identified because the sample correlation between UncDensity and LitDensity is close to zero ($r \approx -0.05$). If the negative association is concentrated in LitDensity while the UncDensity coefficient is small and insignificant, the relation comes mainly from legal and regulatory terminology rather than general uncertainty language.

3.5 H3 — Peer-relative omission gap

H3 asks whether what a firm *leaves out* of Item 1A relative to its industry peers carries information about future volatility, beyond what length and density already capture. Throughout the thesis the *omission gap* is the measure (the share of peer vocabulary a firm leaves out), and the *strategic-silence hypothesis* is the interpretation tested by H3: that a larger omission gap reflects a firm staying quiet on risks its peers disclose.

Construction. For each fiscal year t we build a TF–IDF representation of every Item 1A in the panel, keeping terms that appear in at least five filings and at most 95% of filings, with ℓ_2-normalised rows. Let $\mathbf{x}_{i,t}$ be the TF–IDF row for firm i in year t. For each two-digit SIC industry j in year t with at least six panel filings, the leave-one-out peer aggregate is

$$\mathbf{p}_{i,t} \;=\; \sum_{k \in S_{j(i),t}, k \neq i} \mathbf{x}_{k,t}, \tag{3.8}$$

where $S_{j,t}$ is the set of panel firms in SIC2 industry j that filed in year t. The peer aggregate summarises what the focal firm's industry peers chose to write that year. The omission gap is the share of peer disclosure mass on terms the focal firm does not mention:

$$\textsc{Omission}^{1A}_{i,t} \;=\; 1 \;-\; \frac{\sum_{w:x_{i,t,w}>0} p_{i,t,w}}{\sum_{w} p_{i,t,w}}, \tag{3.9}$$

with values in $[0, 1]$. Zero means the focal firm covers every term any peer mentions. Values near one mean the firm covers only a small share of what its peers chose to discuss. Firm-years in SIC2-year cells with fewer than five peers are dropped because the peer aggregate is too noisy.

Why SIC2 and why control for length. SIC2 is the standard peer benchmark in the disclosure-similarity literature (Brown & Tucker, 2011; Dyer et al., 2017). The median peer cell has 35 firms (P25 = 14). The omission gap is also negatively correlated with Item 1A length ($r = -0.85$), since a longer section mechanically covers more peer terms. The H3 regression controls for $\ln(\text{Words}^{1A})$, so the omission coefficient holds length fixed. The H3 control vector also replaces asset growth with two liquidity proxies, the current ratio and operating cash flow over assets.[1]

Specification. The H3 regression adds the omission gap to the joint specification:

$$\begin{aligned}\ln \sigma^{[h]}_{i,t+1} \;=\; \alpha \;+\; \phi \ln \sigma^{[h]}_{i,t} \;+\; \boldsymbol{\beta}' \mathbf{X}^{\text{fin}}_{i,t} \;+\; \beta^{\text{len}} \cdot \ln(\text{Words}^{1A}_{i,t}) \\ +\; \beta^{\text{dens}} \cdot \text{RiskDensity}_{i,t} \;+\; \beta^{\text{om}} \cdot \textsc{Omission}^{1A}_{i,t} \;+\; \mu_j \;+\; \tau_t \;+\; \varepsilon_{i,t}.\end{aligned} \tag{3.10}$$

Equation 3.10 is run under two fixed-effect structures: the cross-sectional version with μ_j a SIC2 industry fixed effect, and a stricter within-firm version in which μ_j is replaced by a firm fixed effect μ_i. The within-firm version is more demanding because it uses only year-to-year variation in omission driven by changes in peer disclosure. H3 is *fully* supported if $\beta^{\text{om}} > 0$ at the headline $h = 30$ days under both structures. If the sign is positive and significant only under the stricter firm-FE structure, H3 receives *partial* support: the strategic-silence channel is then identified within firms but not as a between-firm regularity. Horizons run over $h \in \{5, 10, 30, 90, 180\}$ days. $h = 365$ days is dropped because the post-filing window for fiscal years 2023–2024 extends past the data cut-off.

The H3 sample has 4,857 to 4,871 firm-years at $h \geq 10$ days, smaller than the H1+H2 sample of 6,048 at $h = 30$ days. There are two reasons: the ≥ 5 peers-per-cell restriction, and the two liquidity controls, which are missing for some firm-years. At $h = 5$ days the sample falls to 4,226, because many 5-day windows do not contain enough daily returns.

[1] The swap follows prior risk-disclosure studies that include liquidity-related controls (Dyer et al., 2017; Kravet & Muslu, 2013). Section 5.4 re-estimates equation 3.10 with the baseline H1+H2 control vector to confirm that the H3 result is not driven by the choice of controls.

3.6 Estimation and inference

All regressions are estimated by ordinary least squares with the relevant fixed effects absorbed using the `linearmodels` package. Coefficients are identified from *within-industry, within-year* (or within-firm, within-year) variation, which removes common time trends in volatility (such as the 2008 and 2020 crises) and persistent industry- or firm-level differences in risk vocabulary. The analysis is predictive, not causal: the coefficients measure the conditional association between Item 1A disclosure and *subsequent* realised volatility, as in Campbell et al. (2014), not the effect of disclosure on volatility.

Limitation: anticipated volatility. Item 1A is written when management already knows current volatility conditions, so expected volatility could shape the wording (reverse causality). The design addresses this only partly: the outcome window starts two trading days after filing, lagged log-volatility is controlled for, fixed effects absorb persistent firm and industry differences, and the lead-lag check in Section 5.4 tests the timing. It does not rule out that management anticipates *future* volatility not captured by lagged volatility, so we interpret the results as predictive associations.

Primary horizon. The headline horizon is $h = 30$ days, at which H1–H3 are tested. The other horizons in Table 5.1 are reported as exploratory evidence on the time decay of the signal.

Standard errors. Standard errors are two-way clustered by firm and fiscal year (Cameron et al., 2011; Petersen, 2009), because (i) consecutive firm-year volatility windows overlap, which creates within-firm serial correlation, and (ii) calendar-year shocks (above all COVID in 2020) create within-year cross-sectional correlation that one-way firm clustering would miss.

Outliers. Financial ratios are winsorised at the 1st and 99th percentiles of the regression sample. The text variables are not winsorised: $\ln(\text{Words}^{1A})$ is bounded by the log transform and RiskDensity is a ratio with sample maximum ≈ 0.05.

Sample. The estimation sample at $h = 30$ days has 6,048 firm-years on 526 unique firms; the $h = 365$ days sample has 6,038 firm-years on the same firms. The 5-day sample is smaller (5,234 firm-years), because many 5-day windows do not contain enough daily returns.

Limitation: survivorship. The list of firms comes from the SEC company list as it is today, so it only contains firms that still exist. Firms that were delisted, acquired, or went bankrupt during 2010–2024 are missing, and the 90% return-coverage rule (Section 2.1) removes a few

more. The panel grows from 149 firms in fiscal year 2010 to over 500 in 2020–2024, which shows this. Missing firms are likely to be riskier, so the results describe large firms that survived, and they may not hold for firms in distress.

Stationarity. Because the specification is dynamic—lagged log-volatility enters with its own coefficient ϕ—an integrated regressor could produce a spurious association. We test every regression variable for a unit root with the augmented Dickey–Fuller test. The results, in Section 5.5, indicate that the panel is stationary, $I(0)$.

4. Results

This chapter reports the empirical results.

4.1 Descriptive statistics

Table 4.1 reports summary statistics for the 30-day regression sample. The dependent variable has mean $\ln \sigma_{i,t+1}^{[30]} \approx -1.34$, corresponding to a typical annualised post-filing volatility around 0.26. Median Item 1A length is $\exp(9.063) \approx 8{,}600$ words. The mean RiskDensity of 0.051 implies that about one in every 20 words in Item 1A is a risk word. The LM-uncertainty sub-list is about twice as dense as the LM-litigious sub-list.

Table 4.1: Summary Statistics

	Mean	SD	P25	Median	P75
$\ln(\sigma_{i,t+1})$	-1.341	0.582	-1.733	-1.408	-1.060
$\ln(\sigma_{i,t})$	-1.352	0.590	-1.754	-1.421	-1.063
ln(Total Assets)	23.139	1.480	22.133	23.163	24.214
Leverage	0.612	0.219	0.471	0.608	0.737
ROA	0.062	0.089	0.027	0.062	0.105
Asset Growth	0.117	0.249	0.003	0.060	0.145
$\ln(\text{Words}^{1A})$	9.011	0.653	8.610	9.063	9.414
RiskDensity	0.051	0.007	0.047	0.051	0.055
UncDensity	0.034	0.005	0.032	0.034	0.037
LitDensity	0.016	0.005	0.013	0.016	0.019
Observations			6,048		
Firms			526		

Note: Summary statistics for the regression sample at the 30-day horizon. $\sigma_{i,t+1}$ is the annualised standard deviation of daily log returns over the 30-day window starting two trading days after the filing date; $\ln(\sigma_{i,t+1})$ is its natural logarithm. Words^{1A} is the cleaned word count of Item 1A; RiskDensity is the share of LM uncertainty- and litigious-list words in Item 1A; UncDensity and LitDensity are the two sub-components separately. Financial ratios are winsorised at the 1st and 99th percentiles.

Figure 4.1 plots the median annualised 30-day post-filing volatility by fiscal year, together with the inter-quartile band. Volatility spikes in 2020 and is otherwise fairly stable. The fiscal-year fixed effects remove these common shocks, and they motivate the pre-COVID / COVID-era split in Section 5.2.

Table 4.2 reports pairwise correlations. The two text variables are correlated at $r = -0.30$: longer Item 1A sections tend to have lower risk-word density, consistent with length being driven by firm-specific text rather than by extra generic risk vocabulary. The correlation is low enough for both variables to be separately identified in the joint specification.

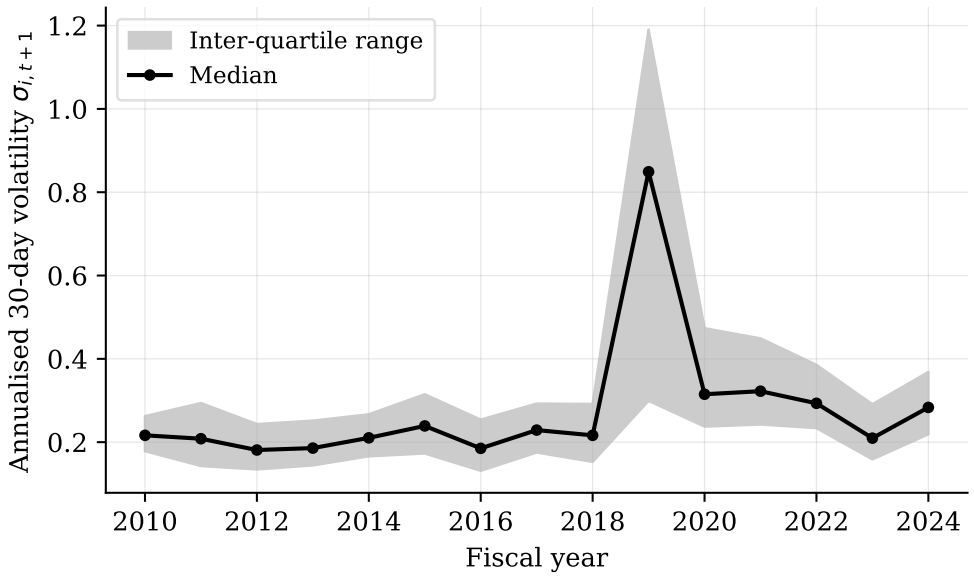


Figure 4.1: Median annualised 30-day post-filing volatility $\sigma_{i,t+1}$ by fiscal year, with the inter-quartile range (shaded). The 2020 spike is the market-wide shock absorbed by the year fixed effects.

Table 4.2: Pearson Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) $\ln(\sigma_{i,t+1})$	1									
(2) $\ln(\sigma_{i,t})$	0.36***	1								
(3) ln(Total Assets)	-0.20***	-0.21***	1							
(4) Leverage	-0.06***	-0.07***	0.21***	1						
(5) ROA	-0.24***	-0.27***	0.08***	-0.05***	1					
(6) Asset Growth	0.17***	0.09***	-0.14***	-0.12***	-0.05***	1				
(7) $\ln(\text{Words}^{1A})$	0.31***	0.32***	-0.09***	-0.03**	-0.28***	0.17***	1			
(8) RiskDensity	-0.11***	-0.12***	0.00	-0.01	0.06***	-0.07***	-0.30***	1		
(9) UncDensity	-0.10***	-0.09***	-0.17***	-0.06***	0.14***	-0.05***	-0.43***	0.71***	1	
(10) LitDensity	-0.06***	-0.08***	0.19***	0.05***	-0.06***	-0.06***	0.04***	0.66***	-0.06***	1

Note: Pairwise Pearson correlation coefficients on the 30-day-horizon regression sample. ***, **, * denote significance at the 1%, 5%, and 10% levels.

4.2 Volume, specificity, and the LM-list decomposition

Table 4.3 reports five nested specifications at the 30-day horizon: financial baseline (1), length only (2), density only (3), the joint H1+H2 specification (4), and the LM-list decomposition (5).

In the joint specification (Column 4), longer Item 1A sections predict higher post-filing volatility ($\hat{\beta}^{\text{len}} = +0.116$, $t = +7.26$) and, holding length fixed, denser sections predict lower volatility ($\hat{\beta}^{\text{risk}} = -4.36$, $t = -3.17$). The density coefficient roughly halves between Columns 3 and 4 (from -8.09 to -4.36): about half of the apparent "risk-density effect" in the dictionary-only specification is the volume effect working through the word-count denominator. **H1 and H2 are both supported.**

To see how big these effects are, note that the dependent variable is in logs, so a coefficient β and a change Δx in the regressor give a change of about $100\,(e^{\beta \Delta x} - 1)\%$ in volatility. For length, one standard deviation (0.653 log points, roughly a 90% longer section) raises volatility by about 8%. A smaller 10% increase in length (0.095 log points) raises it by about 1.1%. At the typical volatility of 0.26, the one-SD effect is about 2 percentage points of annualised volatility. For density, a one-SD rise in RiskDensity (0.007) lowers volatility by about 3.0%. In the decomposition (Column 5), a one-SD rise in LitDensity (0.005) lowers it by about 3.2%. The length effect is the larger of the two; the density effects are smaller.

When the composite RiskDensity is replaced by its two LM sub-densities (Column 5), the negative association comes almost entirely from the LM-Litigious sub-list: $\hat{\beta}^{\text{lit}} = -6.55$ ($t = -3.83$, $p < 0.001$), while $\hat{\beta}^{\text{unc}} = -1.88$ is not statistically significant. A Wald test of equality of the two coefficients gives $z = +1.86$ ($p = 0.063$). So the two coefficients are different only at the 10% level, not at the 5% level, and the claim that the two sub-lists behave differently should be read with care. The pattern is consistent with Beatty et al. (2019) and Cazier et al. (2021): litigious-list language is standardised cautionary boilerplate, and once separated from the uncertainty sub-list it accounts for most of the H2 result. The H2 result is a between-firm pattern: firms that *systematically* use more litigious vocabulary have lower subsequent volatility. It does not mean that a firm which *increases* its litigious language has lower volatility, because both sub-densities lose significance under firm fixed effects (Section 5.3). Adjusted R^2 rises only slightly across the five columns ($0.498 \rightarrow 0.513$) because the financial baseline already explains most of the predictable volatility. Table 4.4 reproduces the same five columns with the dependent variable replaced by $\ln \sigma^{[365]}_{i,t+1}$. The signs are unchanged and the magnitudes are about half of the 30-day estimates.

Table 4.3: Item 1A Length, Risk-Word Density, and the Boilerplate Decomposition (30 days window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\sigma_{i,t})$	+0.1735**	+0.1405*	+0.1579**	+0.1366*	+0.1353*
	(+2.21)	(+1.87)	(+2.03)	(+1.82)	(+1.81)
ln(Total Assets)	−0.0713***	−0.0689***	−0.0709***	−0.0690***	−0.0665***
	(−6.55)	(−6.70)	(−6.80)	(−6.81)	(−6.43)
Leverage	−0.0241	−0.0054	−0.0288	−0.0105	−0.0115
	(−0.75)	(−0.17)	(−0.91)	(−0.33)	(−0.36)
ROA	−1.1053***	−0.9349***	−1.1041***	−0.9577***	−0.9635***
	(−10.83)	(−10.10)	(−10.96)	(−10.23)	(−10.23)
Asset Growth	+0.2381***	+0.1963***	+0.2238***	+0.1943***	+0.1917***
	(+4.55)	(+4.03)	(+4.42)	(+3.99)	(+3.93)
$\ln(\text{Words}^{1A})$		+0.1342***		+0.1158***	+0.1243***
		(+8.14)		(+7.26)	(+7.07)
RiskDensity			−8.0914***	−4.3588***	
			(−5.75)	(−3.17)	
UncDensity					−1.8778
					(−0.93)
LitDensity					−6.5541***
					(−3.83)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	6,048	6,048	6,048	6,048	6,048
Firms	526	526	526	526	526
Adj. R^2	0.498	0.511	0.504	0.512	0.513

Note: The dependent variable is $\ln(\sigma_{i,t+1})$, the natural log of annualised post-filing volatility over a 30 days window starting two trading days after the 10-K filing date. Column (1) reports the financial baseline; column (2) adds the log word count of Item 1A (H1); column (3) adds the composite LM risk-word density (H2); column (4) adds H1 and H2 jointly; column (5) replaces the H2 composite by its LM-list decomposition into the LM-Uncertainty and LM-Litigious densities of Item 1A. All specifications include two-digit SIC industry and fiscal-year fixed effects. t-statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Table 4.4: Item 1A Length, Risk-Word Density, and the Boilerplate Decomposition (365 days window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\sigma_{i,t})$	+0.6035***	+0.5630***	+0.5850***	+0.5579***	+0.5556***
	(+13.58)	(+11.86)	(+12.63)	(+11.60)	(+11.53)
ln(Total Assets)	−0.0343***	−0.0354***	−0.0351***	−0.0357***	−0.0344***
	(−4.73)	(−4.94)	(−4.90)	(−5.01)	(−4.76)
Leverage	−0.0146	−0.0052	−0.0171	−0.0079	−0.0085
	(−1.03)	(−0.35)	(−1.16)	(−0.52)	(−0.55)
ROA	−0.4572***	−0.4018***	−0.4713***	−0.4170***	−0.4221***
	(−7.21)	(−6.91)	(−7.31)	(−7.13)	(−7.13)
Asset Growth	+0.0740**	+0.0582**	+0.0695**	+0.0578**	+0.0565**
	(+2.35)	(+2.13)	(+2.25)	(+2.10)	(+2.05)
$\ln(\text{Words}^{1A})$		+0.0691***		+0.0601***	+0.0652***
		(+5.37)		(+4.78)	(+5.17)
RiskDensity			−4.0296***	−2.2617***	
			(−5.57)	(−3.65)	
UncDensity					−0.8748
					(−1.05)
LitDensity					−3.5079***
					(−3.95)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	6,038	6,038	6,038	6,038	6,038
Firms	526	526	526	526	526
Adj. R^2	0.727	0.733	0.730	0.734	0.734

Note: The dependent variable is $\ln(\sigma_{i,t+1})$, the natural log of annualised post-filing volatility over a 365 days window starting two trading days after the 10-K filing date. Column (1) reports the financial baseline; column (2) adds the log word count of Item 1A (H1); column (3) adds the composite LM risk-word density (H2); column (4) adds H1 and H2 jointly; column (5) replaces the H2 composite by its LM-list decomposition into the LM-Uncertainty and LM-Litigious densities of Item 1A. All specifications include two-digit SIC industry and fiscal-year fixed effects. t-statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

4.3 Strategic-silence hypothesis and within-firm post-filing volatility

Table 4.5 reports the H3 specification of equation 3.10 at five post-filing horizons under two fixed-effect structures: industry plus year FE in Panel A and firm plus year FE in Panel B. The coefficient of interest is β^{om} on the peer-relative omission gap; Item 1A length, RiskDensity, the five financial controls, and lagged log-volatility are partialled out.

Table 4.5: Silence Hypothesis: Item 1A omission relative to SIC2 peers and post-filing realised volatility.

	$h = 5$	$h = 10$	$h = 30$	$h = 90$	$h = 180$
Panel A. Industry × year fixed effects					
Omission_{1A}	+0.256	+0.207	+0.238	+0.272	+0.283
	(+0.41)	(+0.44)	(+0.65)	(+0.95)	(+1.28)
N	4,226	4,857	4,871	4,871	4,871
R^2	0.230	0.372	0.514	0.586	0.655
Panel B. Firm × year fixed effects					
Omission_{1A}	+1.735**	+1.395*	+1.247***	+1.339***	+1.237***
	(+2.05)	(+1.75)	(+2.58)	(+2.77)	(+2.76)
N	4,226	4,857	4,871	4,871	4,871
R^2	0.363	0.492	0.627	0.694	0.750

Notes. Dependent variable is $\log(\text{RV}_{hd})$ over the h-day window beginning two trading days after the filing date. Omission_{1A} is the share of TF-IDF mass on terms used by SIC2-year peers (leave-one-out, ≥ 5 peers) that the focal firm does not mention. All regressions control for $\log(\text{RV}_{hd,t-1})$, the five financial controls (size, leverage, ROA, current ratio, OCF/assets), $\log(\text{words}_{1A})$, and RiskDensity_{1A}. t-statistics in parentheses use two-way clustered standard errors on (ticker, fiscal year). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The cross-section in Panel A is not significant at any horizon ($|t| < 1.3$). In the firm-FE specification of Panel B the coefficient is positive and significant at the 30-day horizon ($\hat{\beta}^{\text{om}} = +1.247$, $t = +2.58$, $p = 0.010$) and also at 90 and 180 days. It is significant at the 5% level at 5 days and only at the 10% level at 10 days. A one-within-firm-SD rise in Omission^{1A} (0.043) implies a $\approx 5.5\%$ increase in 30-day annualised volatility, close to the size of the H1 length effect. One interpretation is that persistent firm-level disclosure styles absorb the cross-sectional variation, while the peer-relative omission gap, which by construction varies with what *other* firms write each year, remains after firm fixed effects. Because the pre-specified criterion in Chapter 3 required a positive β^{om} under *both* structures, and Panel A is positive (+0.238) but not different from zero ($t = +0.65$), **H3 receives only partial support: it is supported within-firm at the 30-day horizon but not in the industry-FE cross-section**. Section 5.4 shows that the within-firm result is also weaker in some robustness checks.

5. Robustness

This chapter checks that the headline results of Chapter 4 survive several variations in the specification.

5.1 Alternative post-filing windows

The headline specification measures post-filing volatility over a 30-day window. To check the choice of window we re-estimate the H1+H2 specification of equation 3.7 at five additional horizons (5, 10, 90, 180, and 365 days). Table 5.1 reports all six columns side by side.

Table 5.1: Post-Filing Volatility at Alternative Horizons

	(1) 5d	(2) 10d	(3) 30d	(4) 90d	(5) 180d	(6) 365d
$\ln(\sigma_{i,t})$	+0.0417*	+0.1173***	+0.1353*	+0.2187**	+0.3549***	+0.5556***
	(+1.73)	(+3.79)	(+1.81)	(+2.38)	(+4.35)	(+11.53)
$\ln(\text{Words}^{1A})$	+0.1697***	+0.1415***	+0.1243***	+0.1130***	+0.0937***	+0.0652***
	(+5.67)	(+7.20)	(+7.07)	(+6.09)	(+5.18)	(+5.17)
UncDensity	−4.2493	−1.5255	−1.8778	−0.8084	−0.7444	−0.8748
	(−1.21)	(−0.72)	(−0.93)	(−0.47)	(−0.58)	(−1.05)
LitDensity	−4.1523	−6.6835***	−6.5541***	−6.7992***	−5.2998***	−3.5079***
	(−1.36)	(−2.76)	(−3.83)	(−4.13)	(−4.19)	(−3.95)
Financial controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,234	6,032	6,048	6,048	6,048	6,038
Firms	523	526	526	526	526	526
Adj. R^2	0.216	0.367	0.513	0.583	0.645	0.734

Note: Each column re-estimates the H1 + LM-list decomposition specification replacing the dependent variable with $\ln(\sigma^{[h]}_{i,t+1})$ measured over a post-filing window of length h. The lagged dependent variable is the same firm's log-volatility computed over the prior filing year's window of identical length. UncDensity and LitDensity are the LM-Uncertainty and LM-Litigious word densities on Item 1A; together they sum to RiskDensity (H2). Financial controls (size, leverage, ROA, asset growth) are included but suppressed. Industry and year fixed effects are absorbed; t-statistics in parentheses use two-way (firm and fiscal-year) clustered standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels.

All three text coefficients keep the predicted sign at every horizon: positive for $\ln(\text{Words}^{1A})$, negative for LitDensity, and not significantly different from zero for UncDensity. Neither H1 nor the LM-list decomposition reverses for any of the windows. The length coefficient is significant at the 1% level at every horizon. The LitDensity coefficient is significant at the 1% level from 10 days onward, with $|t|$ between 2.76 (10 days) and 4.19 (180 days). At 5 days it has the same sign but is not significant (-4.15, $t = -1.36$). A 5-day window holds only

a few daily returns, so the volatility estimate is noisy and the sample is smaller (5,234 firm-years). The UncDensity coefficient is never significant ($|t| \leq 1.21$ at every window). The 30-day window is not one in which the results happen to be strongest.

Both coefficient series decline in a similar way with h, consistent with a signal that is priced over weeks rather than years. We do not fit a formal half-life because the error terms across horizons are correlated through the shared daily returns.

Figure 5.1 plots the two headline coefficients with their 95% confidence intervals across all six horizons. The length effect (left) is positive and significant throughout, and the litigious-density effect (right) is negative throughout and significant from 10 days onward. Both become smaller at longer windows.

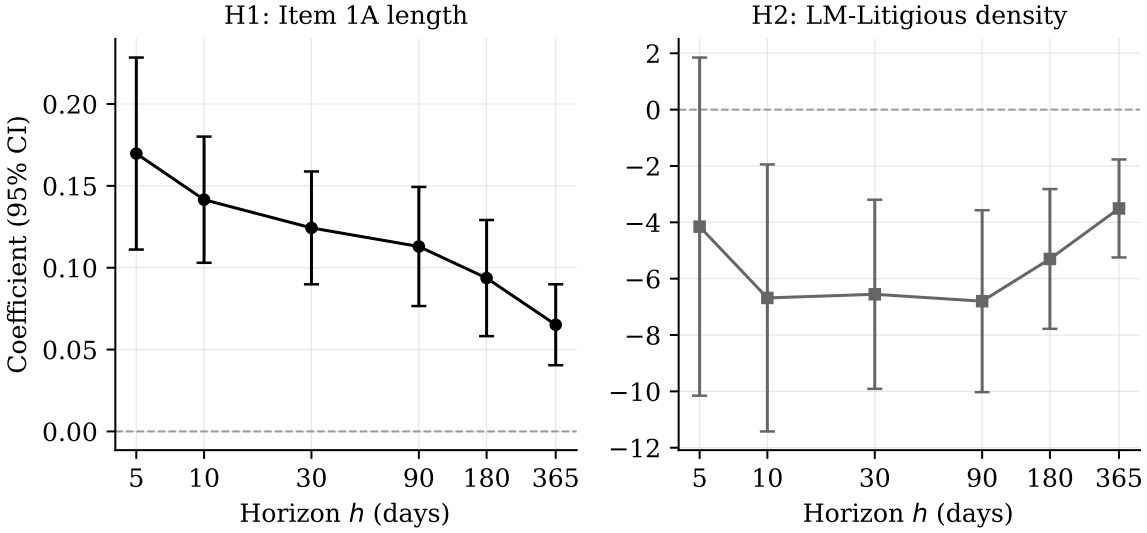


Figure 5.1: Length (H1, left) and LM-Litigious density (H2, right) coefficients with 95% confidence intervals across the six post-filing horizons ($h = 5$, 10, 30, 90, 180, and 365 days; horizons are displayed at equal spacing for readability). Both effects keep their sign at every horizon and decay towards longer windows.

5.2 Pre-COVID and COVID-era sub-samples

The 2020 fiscal year and its aftermath were a large shock to both equity volatility and corporate disclosure. Average annualised volatility roughly doubled in spring 2020, and Item 1A sections grew longer as firms added pandemic and supply-chain language. We re-estimate the H1+H2 decomposition specification (equation 3.7) separately for the two periods. Filings with fiscal year ≤ 2019 form the pre-COVID sample (3,595 firm-years) and filings with fiscal year ≥ 2020 form the COVID-era sample (2,453 firm-years). All other features of the specification are unchanged. Table 5.2 reports the result.

Length and LitDensity have the predicted signs and similar magnitudes in both periods.

Table 5.2: Sub-Period Robustness (30-day horizon)

	(1) Pre-COVID (2010–2019)	(2) COVID era (2020–2024)
$\ln(\sigma_{i,t})$	+0.1433	+0.1221
	(+1.39)	(+1.63)
$\ln(\text{Words}^{1A})$	+0.0997***	+0.1682***
	(+5.99)	(+8.58)
UncDensity	−3.0213	+0.1153
	(−1.20)	(+0.04)
LitDensity	−5.8641***	−10.1748***
	(−2.74)	(−5.64)
Financial controls	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	3,595	2,453
Firms	474	519
Adj. R^2	0.530	0.440

Note: H1 + LM-list decomposition specification re-estimated on two non-overlapping sub-samples. The dependent variable is $\ln(\sigma^{[30]}_{i,t+1})$. UncDensity and LitDensity are the LM-Uncertainty and LM-Litigious word densities on Item 1A. All financial controls and SIC2+year fixed effects are included. Standard errors are two-way clustered by firm and fiscal year. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Length is significant at the 1% level in both ($\beta = +0.100$, $t = +5.99$ pre-COVID; $\beta = +0.168$, $t = +8.58$ COVID era). LitDensity is negative and significant in both ($\beta = -5.86$, $t = -2.74$ pre-COVID; $\beta = -10.18$, $t = -5.64$ COVID era). UncDensity is insignificant in both, as in the full sample. Length and LitDensity coefficients are larger in absolute value in the COVID-era sample, consistent with the higher level of volatility in that period, but the pattern is the same. The main results are not driven by COVID.

5.3 Within-firm identification: firm fixed effects

The headline specification absorbs two-digit SIC industry and fiscal-year fixed effects, so coefficients come from variation *within an industry-year cell*. This removes industry differences in risk vocabulary and macroeconomic time effects, but persistent firm-level traits could still drive the result. We replace the industry fixed effects with firm fixed effects, so that identification comes only from *within-firm* year-to-year variation in the three text variables. Table 5.3 reports the result at the 30-day horizon. Standard errors are clustered by both firm and fiscal year.

Table 5.3: Within-Firm Identification (Firm + Year FE, 30-day horizon)

	Estimate
$\ln(\sigma_{i,t})$	-0.0867
	(-1.21)
$\ln(\text{Words}^{1A})$	$+0.0693^{**}$
	$(+2.57)$
UncDensity	-2.3919
	(-0.92)
LitDensity	-1.0387
	(-0.26)
Financial controls	Yes
Firm FE	Yes
Year FE	Yes
Observations	6,048
Firms	526
Adj. R^2	0.581

Note: H1 + LM-list decomposition specification with firm and fiscal-year fixed effects absorbed. Identification of $\ln(\text{Words}^{1A})$, UncDensity and LitDensity comes only from within-firm variation across years. Standard errors are two-way clustered by firm and fiscal year. ***, **, * denote significance at the 1%, 5%, and 10% levels.

The length effect remains, though it is smaller: $\hat{\beta}^{\text{len}} = +0.069$ ($t = +2.57$, $p \approx 0.01$). Firms that lengthen their Item 1A from one year to the next have higher subsequent volatility. Both LM sub-densities become insignificant: $\hat{\beta}^{\text{unc}} = -2.39$ ($t = -0.92$) and $\hat{\beta}^{\text{lit}} = -1.04$

($t = -0.26$). The mix of risk vocabulary changes little within firms because Item 1A is largely carried over from one filing to the next (Cazier et al., 2021), so there is little variation left to estimate a sub-density coefficient from. This fits the Beatty et al. (2019) and Cazier et al. (2021) view of cautionary boilerplate as a near-permanent firm-level disclosure style. We therefore interpret the H2 result as a between-firm pattern and do not claim a within-firm causal channel.

Figure 5.2 shows the contrast. Under industry and year fixed effects the length coefficient is positive, both density coefficients are precisely estimated, and LitDensity is below zero. With firm and year fixed effects the length effect is almost unchanged, but the density intervals widen and both include zero.

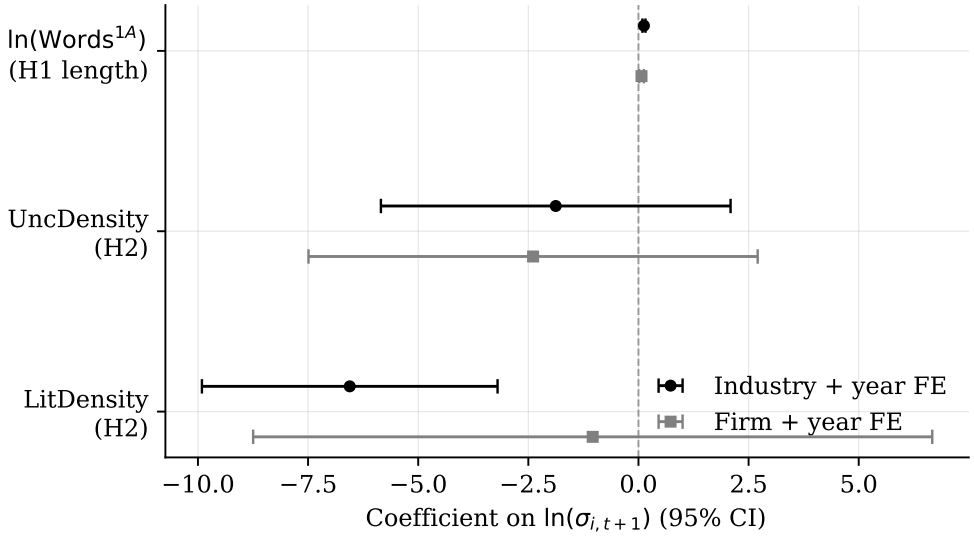


Figure 5.2: H1 and H2 coefficients at the 30-day horizon with 95% confidence intervals, under industry+year fixed effects (circles) versus firm+year fixed effects (squares). The length effect is stable across the two designs; the density coefficients lose significance once firm fixed effects absorb permanent disclosure styles.

5.4 Robustness of the strategic-silence result (H3)

The H3 result of Section 4.3 comes from a within-firm specification at the 30-day horizon. Three features of the design call for further checks before reading it as evidence of strategic silence: (i) the omission gap is mechanically negatively correlated with Item 1A length ($r = -0.93$), so with a weak length control the omission gap could proxy for length; (ii) the SIC2 peer benchmark is one of several possible reference sets, and the result might depend on that choice; (iii) the filing date and the start of the volatility window are close together, which leaves room for reverse causality if a difficult year produces both a shorter, less peer-aligned 10-K and higher volatility for unrelated reasons. Table 5.4 reports six specifications that address these concerns; a placebo test is reported separately below.

Figure 5.3 illustrates the first concern. Longer Item 1A sections cover more of the peer vocabulary, so the omission gap falls steeply with length ($r = -0.93$). Column (3) of Table 5.4, which removes the length component from the omission gap by orthogonalising it on $\ln(\text{WORDS}_{1A})$, is therefore the key check for H3.

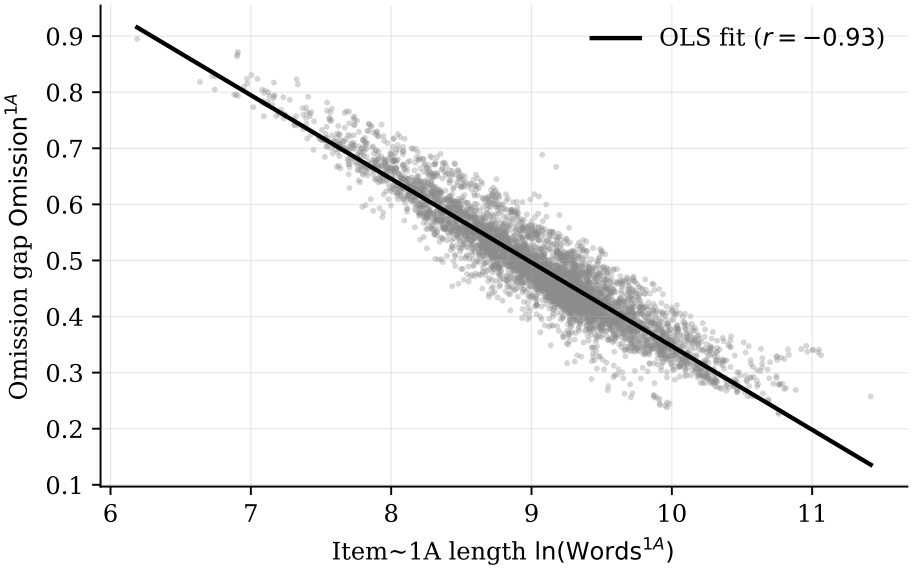


Figure 5.3: Item 1A omission gap against Item 1A length, one point per firm-year, with the OLS fit line. The strong negative association ($r = -0.93$) is the central identification concern for H3 and motivates the length-orthogonalised specification in Column (3) of Table 5.4. Firm-years with non-recoverable Item 1A text ($\text{WORDS}_{1A} = 0$, $n = 63$) are excluded from the figure: these observations mechanically take $\text{OMISSION}_{1A} = 1$ and do not enter any regression because the LM-density variables are undefined for them.

Placebo (peer-cell shuffle). The first check keeps the SIC2 cell sizes and the firm-year distribution fixed but randomly reassigns firms to SIC2 cells within each fiscal year, recomputes the omission gap of equation 3.9, and re-estimates equation 3.10 with firm and year fixed effects. If the association reflects silence on *peer-relevant* risks, the placebo coefficient should disappear. If it is a length artefact, the placebo should resemble the real coefficient. Across five permutations the placebo t-values are $1.00, 0.59, 2.19, 1.20, 1.39$. Four of the five are small, but one ($t = 2.19$) is significant at the 5% level. The real coefficient ($t = +2.58$) is only a little above the largest placebo value. So the placebo test gives only weak support: a random peer group can sometimes produce a similar association, most likely because the omission gap still carries some length information.

Length controls (Columns 3 and 5). Column (3) replaces the omission gap by its residual after regressing on $\ln(\text{WORDS}_{1A})$ within each (SIC2, fiscal year) cell; the residual has the predicted sign but is not significant ($t = +1.54$, $p = 0.12$). Column (5) keeps the original gap but adds two more length controls (MD&A word count and squared Item 1A log word count); the omission coefficient stays significant ($t = +2.85$, $p = 0.004$). Linear or quadratic

Table 5.4: Robustness of the strategic-silence result (H3): firm × year fixed effects, $h = 30$ days.

	(1) Baseline	(2) Decile rank	(3) Length-resid	(4) ≥ 20 peers	(5) Triple length	(6) Prior-year
Omission_{1A}	+1.247***	+0.006	+0.310	+2.470***	+1.360***	+0.448*
	(+2.58)	(+0.74)	(+1.54)	(+3.01)	(+2.85)	(+1.88)
N	4,871	3,876	4,867	3,396	4,871	4,447

Notes. Each column is a separate firm × year fixed-effect regression at $h = 30$ days. Column (1) reports the baseline H3 specification of equation 3.10. Column (2) replaces Omission_{1A} with its decile rank within (SIC2, fiscal year) cells. Column (3) replaces it with the residual after regressing Omission_{1A} on $\ln(\text{Words}_{1A})$ within (SIC2, fiscal year) cells. Column (4) restricts the sample to industry-year cells with at least 20 panel firms. Column (5) adds $\ln(\text{Words}_{\text{MDA}})$ and $\ln(\text{Words}_{1A})^2$ as additional length controls. Column (6) replaces the contemporaneous omission with its prior-year value. All regressions include lagged log-volatility, the five financial controls, $\ln(\text{Words}_{1A})$, and RiskDensity_{1A}. t-statistics in parentheses use two-way clustered standard errors on (ticker, fiscal year). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

length controls do not remove the effect, but once the length part of the gap is fully removed (Column 3) the effect is no longer significant.

Large-cell restriction (Column 4) and prior-year specification (Column 6). Restricting to cells with at least 20 peers reduces the panel to 3,396 firm-years and *increases* the coefficient to +2.47 ($t = +3.01$), so the effect is stronger where the peer aggregate is measured more precisely. Column (6) uses prior-year omission to address the filing-window overlap; the lagged coefficient is positive but significant only at the 10% level (+0.448, $t = +1.88$, $p = 0.06$).

Decile rank (Column 2). Using the decile rank of omission within each (SIC2, fiscal year) cell keeps the positive sign but is not significant ($t = +0.74$). If there is an effect, it sits in the tails of the omission distribution, where omissions are most visible.

Summary for H3. The omission coefficient has the predicted positive sign in every column. It stays significant with extra length controls and in the large-cell sample. However, it is not significant once the length part is fully removed (Column 3) or with the decile rank (Column 2), it is only significant at the 10% level with prior-year omission (Column 6), and one of five placebo runs comes close to the real t-value. We therefore treat the H3 result as suggestive only: it is consistent with a within-firm strategic-silence channel, but we cannot rule out that it is partly a length effect, and we make no causal claim.

5.5 Stationarity of the panel

Every specification in this thesis is a dynamic panel: lagged log-volatility enters equation 3.3 and its extensions with its own coefficient ϕ. If the outcome or a regressor had a unit root, the regression could be spurious. We test all twelve regression variables for stationarity with the (augmented) Dickey–Fuller test.

How the test is applied. Each variable is a panel of short annual series, with at most fifteen observations per firm. We adapt the Dickey and Fuller (1979) test in two steps. First, we subtract each year's cross-sectional mean from every variable, which removes common shocks such as the 2020 volatility spike, as the year fixed effects do in the regressions. Second, we run an augmented Dickey–Fuller regression (intercept, one lag) for each firm and combine the per-firm p-values with Fisher's method (Maddala & Wu, 1999). Pooling is needed because a single fifteen-year series has little power. Table 5.5 reports, for each variable, the average autoregressive coefficient, the average ADF statistic, the share of firms that reject the unit root individually, and Fisher's combined p-value.

Autoregressive coefficients. A unit root means $\rho = 1$. The estimates of $\hat{\rho}$ range from -0.07 for the (demeaned) outcome to at most 0.64, far from one. After removing the common market factor, firm-specific log-volatility is almost serially uncorrelated ($\hat{\rho} \approx 0$), and the most persistent variables (leverage, Item 1A length, and the risk-word densities) have $\hat{\rho}$ below two-thirds.

Per-firm and pooled tests. The average ADF statistic lies between -1.6 and -2.6, short of the -2.9 needed to reject at 5% for a single series, and only 10–30% of firms reject individually. This is expected given fifteen-year series. The pooled Fisher test rejects the unit-root null with a p-value below 0.001 for all twelve variables. The Dickey–Fuller test without the augmentation lag rejects more often (the share of firms rejecting roughly doubles), so the result does not depend on the lag choice.

Bounded variables. The variables with the highest $\hat{\rho}$, leverage and the risk-word densities, are bounded by construction: leverage lies between zero and one, and the densities are word shares with sample maxima below 0.05. A unit-root process has variance that grows without bound, so boundedness is evidence against a unit root, though not a proof of stationarity. Together with the test results above, it makes an integrated process implausible. The persistence of these variables reflects the fact that Item 1A is largely carried over from one filing to the next (Cazier et al., 2021), a slow-moving but mean-reverting firm trait, as in the firm fixed-effects result of Section 5.3.

Table 5.5: Stationarity Tests (Augmented Dickey–Fuller)

	$\hat{\rho}$ (AR1 coef.)	ADF $\bar{t}$ (mean)	Reject (% firms)	Fisher p-value
$\ln \sigma_{i,t}$ (outcome)	−0.07	−2.37	23%	< 0.001
ln(Total Assets)	0.58	−1.87	17%	< 0.001
Leverage	0.64	−1.74	12%	< 0.001
ROA	0.32	−2.03	17%	< 0.001
Asset Growth	0.01	−2.59	29%	< 0.001
Current Ratio	0.36	−2.05	19%	< 0.001
OCF/Assets	0.32	−2.03	16%	< 0.001
ln(Words^{1A})	0.64	−2.23	14%	< 0.001
RiskDensity	0.58	−1.75	12%	< 0.001
UncDensity	0.59	−1.79	11%	< 0.001
LitDensity	0.59	−1.61	10%	< 0.001
Omission^{1A}	0.64	−1.69	10%	< 0.001

Note: Stationarity tests for every variable that enters the regressions of Chapters 4–5; lagged log-volatility is the lag of the outcome and is covered by the first row. Each variable is a panel of short annual firm-level series ($T \leq 15$). Before testing, each year's cross-sectional mean is subtracted to remove the market-wide shocks shared by all firms, mirroring the year fixed effects in the regressions. For every firm with at least 8 annual observations we run an augmented Dickey–Fuller regression (Dickey & Fuller, 1979) with an intercept and one lag. $\hat{\rho}$ is the average estimated first-order autoregressive coefficient across firms; a unit root would imply $\hat{\rho} = 1$, so values well below one indicate mean reversion. ADF $\bar{t}$ is the mean ADF statistic (the 5% critical value with an intercept is about −2.9), and "Reject" is the share of firms rejecting the unit-root null at 5%. Because each firm's series is short and individually low-powered, the last column combines the per-firm ADF p-values across firms with Fisher's method (Maddala & Wu, 1999); the combined p-value is below 0.001 for every variable, so the unit-root null is rejected for all of them. Plain Dickey–Fuller without the augmentation lag gives the same conclusion. The most persistent variables (Leverage and the risk-word densities) are bounded ratios and shares, which cannot follow a unit-root process—whose variance grows without bound—so their persistence is mean reversion, not a unit root. Every variable is therefore treated as stationary, $I(0)$.

Summary. All regression variables have autoregressive coefficients well below one, and the pooled Dickey–Fuller test rejects the unit-root null for all twelve at the 0.1% level. The panel is stationary, $I(0)$, and the main associations are not spurious regressions.

Conclusion

References

Agresti, A., & Coull, B. A. (1998). Approximate is better than "exact" for interval estimation of binomial proportions. *The American Statistician*, *52*(2), 119–126.

Beatty, A., Cheng, L., & Zhang, H. (2019). Are risk factor disclosures still relevant? evidence from market reactions to risk factor disclosures before and after the financial crisis. *Contemporary Accounting Research*, *36*(2), 805–838.

Benoit, K., Conway, D., Lauderdale, B. E., Laver, M., & Mikhaylov, S. (2016). Crowd-sourced text analysis: Reproducible and agile production of political data. *American Political Science Review*, *110*(2), 278–295.

Brown, S. V., & Tucker, J. W. (2011). Large-sample evidence on firms' year-over-year MD&A modifications. *Journal of Accounting Research*, *49*(2), 309–346.

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, *29*(2), 238–249.

Campbell, J. L., Chen, H., Dhaliwal, D. S., Lu, H.-m., & Steele, L. B. (2014). The information content of mandatory risk factor disclosures in corporate filings. *Review of Accounting Studies*, *19*(1), 396–455.

Cazier, R. A., McMullin, J. L., & Treu, J. S. (2021). Are lengthy and boilerplate risk factor disclosures inadequate? an examination of judicial and regulatory assessments of risk factor language. *The Accounting Review*, *96*(4), 131–155.

Chychyla, R., & Kogan, A. (2015). Using XBRL to conduct a large-scale study of discrepancies between the accounting numbers in Compustat and SEC 10-K filings. *Journal of Information Systems*, *29*(1), 37–72. https://doi.org/10.2308/isys-50922

Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, *74*(366), 427–431.

Dyer, T., Lang, M., & Stice-Lawrence, L. (2017). The evolution of 10-K textual disclosure: Evidence from Latent Dirichlet Allocation. *Journal of Accounting and Economics*, *64*(2–3), 221–245.

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, *120*(30), e2305016120.

Hope, O.-K., Hu, D., & Lu, H. (2016). The benefits of specific risk-factor disclosures. *Review of Accounting Studies*, *21*(4), 1005–1045.

Kravet, T., & Muslu, V. (2013). Textual risk disclosures and investors' risk perceptions. *Review of Accounting Studies*, *18*(4), 1088–1122.

Li, F. (2008). Annual report readability, current earnings, and earnings persistence. *Journal of Accounting and Economics*, *45*(2–3), 221–247.

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, *66*(1), 35–65.

Maddala, G. S., & Wu, S. (1999). A comparative study of unit root tests with panel data and a new simple test. *Oxford Bulletin of Economics and Statistics*, *61*(S1), 631–652.

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *The Review of Financial Studies*, *22*(1), 435–480.

Appendices

A. Variable Definitions and Data Pipeline

This appendix collects supplementary tables referenced in the main text. Section A.1 presents the diagnostics behind the choice of annual filing frequency (Section 2.2). Section A.2 reports the per-firm XBRL tag assignments from the tag-harmonisation procedure (Section 2.3).

A.1 Filing Frequency Diagnostics

The following tables supplement the discussion in Section 2.2.

Table A.1: XBRL Financial Metric Coverage: Annual (10-K) vs. Quarterly (10-Q)

Metric	Annual (%)	Quarterly (%)
assets_current	87.0	91.1
eps_diluted	89.4	92.7
liabilities_current	87.0	91.1
net_income	93.6	97.3
operating_cash_flow	89.2	35.8
operating_income	76.9	80.5
stockholders_equity	98.7	94.9
total_assets	89.9	93.3
total_liabilities	63.7	66.5
Median periods per firm	20	52
Firms with data	30	30

Note: Coverage shows the percentage of firm-periods with non-missing values for each metric. Annual data is extracted from 10-K filings; quarterly data from 10-Q filings (Q1–Q3 only). Sample: 30 non-financial operating companies from the top 200 by market capitalisation.

Table A.2: Annual vs. Quarterly Approach: Summary Comparison

Criterion	Annual (10-K)	Quarterly (10-Q)
Filing type	10-K	10-Q (Q1–Q3 only)
MD&A availability	93% (Item 7)	100% (Part I, Item 2)
Risk Factors availability	100% (Item 1A)	∼52% substantive
Risk Factors content	Median ∼30,000 words	Median ∼9,000 words (substantive); 48% boilerplate
Q4 financial data	Included in 10-K	Unreliable / absent
Explicit Q4 XBRL facts	N/A	3–43% of firm-years
Implied Q4 mismatch rate	N/A	∼11% of comparisons
Cash-flow quarterly coverage	N/A	7% of firm-years
Item 1A length (H1)	Feasible (annual word count)	Limited (52% substantive)
Item 1A risk-word density (H2)	Feasible (annual LM ratio)	Limited (52% substantive)
Literature alignment	Standard	Uncommon
Expected observations	∼6,400 firm-years	∼19,000 firm-quarters

Note: Summary based on diagnostic tests of 30 non-financial companies (XBRL, Q4 tests) and 15 companies (textual tests). Risk Factors in 10-Q filings are required by SEC Regulation S-K Item 1A only when there are "material changes" from the most recent 10-K; roughly half of companies include substantive risk disclosures, while the rest provide only a boilerplate reference. See Tables A.1, 2.1, and 2.2 for detailed results.

Table A.3: Per-Company 10-Q Textual Content (Most Recent 6 Filings)

Ticker	Filings	MD&A Found	MD&A Med. Words	Risk Subst.	Risk Boilerplate
AAPL	6	6	2,781	3	3
AMZN	6	6	6,291	6	0
AVGO	6	6	4,778	6	0
COST	6	6	4,979	0	6
GOOGL	6	6	8,096	3	3
JNJ	6	6	7,587	0	6
LLY	6	6	4,994	0	6
MA	6	6	7,300	6	0
META	6	6	7,756	6	0
MSFT	6	6	7,981	0	6
NVDA	6	6	5,156	6	0
TSLA	6	6	5,582	2	4
V	6	6	5,043	0	6
WMT	6	6	7,142	3	3
XOM	6	6	6,197	6	0

Note: "MD&A Found" and "Risk Subst." count filings with >200 substantive words. "Risk Boilerplate" counts filings where the risk-factors section contains only a reference to the annual 10-K.

A.2 XBRL Tag Provenance

Table A.4 reports, for each metric with multiple candidate XBRL tags, the number and share of firms locked to each tag by the per-firm tag-locking procedure described in Section 2.3.

Table A.4: Per-firm XBRL tag distribution

Metric	Locked XBRL tag	Firms	%
Net income	`NetIncomeLoss`	502	91.6
	`ProfitLoss`	45	8.2
	`IncomeLossFromContinuingOperations`	1	0.2
Operating cash flow	`NetCashProvidedByUsedInOperatingActivities`	530	96.7
	`...OperatingActivitiesContinuingOperations`	18	3.3
Stockholders' equity	`StockholdersEquityIncl...NoncontrollingInterest`	318	58.6
	`StockholdersEquity`	225	41.4

Notes. For each (firm, metric) pair, the tag with the highest within-firm year-count is selected; values from other tags are set to missing. Firm counts are based on up to 548 firms in the tag-locking stage; the stockholders' equity row sums to 543 because five firms report no equity tag at all. See Section 2.3 for the rationale.

B. Documentation of AI Tool Usage

This appendix documents the use of artificial intelligence tools in preparing this thesis, as required by the university.

Extraction Validation (Section 2.4.4)

Tool. Google Gemini 2.5 Flash (accessed via the OpenAI-compatible Gemini API endpoint), with temperature set to 0 for deterministic output.

Purpose. An independent, automated quality audit of the SEC 10-K section extraction pipeline. The model evaluated 200 extracted sections (100 Item 1A, 100 Item 7) against a structured five-criterion rubric (correct section, start boundary, end boundary, content quality, completeness) on a three-level scale (Pass / Minor / Fail).

Contribution and verification. The evaluation rubric, sampling strategy, prompt design, and statistical analysis (Wilson confidence intervals) were all designed by the author. The model's only role was applying the rubric to each extracted section. All Fail verdicts were manually spot-checked against the original SEC filings and confirmed to be correct. Results are reported in Table B.1.

Table B.1: LLM-assisted extraction validation results (n = 200 sections, 100 per section type).

Criterion	Strict	%	Lenient	%	95 % CI (lenient)
Correct section	192/200	96.0	192/200	96.0	[92.3 – 98.0 %]
Start boundary	191/200	95.5	193/200	96.5	[93.0 – 98.3 %]
End boundary	192/200	96.0	193/200	96.5	[93.0 – 98.3 %]
Content quality	191/200	95.5	195/200	97.5	[94.3 – 98.9 %]
Completeness	189/200	94.5	191/200	95.5	[91.7 – 97.6 %]
Overall (all five)	179/200	89.5	187/200	93.5	[89.2 – 96.2 %]

Code Development

Tool. AI-powered code completion (integrated in VS Code).

Purpose. Code completion during the development of the data-pipeline scripts that build the panel and extract the 10-K text sections.

Contribution and verification. All generated code was reviewed, tested, and modified by the author. The overall architecture, variable definitions, and analytical choices are the author's own.

Writing Assistance

Tool. AI-powered writing assistant (integrated in VS Code).

Purpose. Grammar and style suggestions for English-language prose in the thesis text.

Contribution and verification. All substantive content, argumentation, and interpretation are the author's own. The tool was used only for language polishing; every suggestion was reviewed before acceptance.